chronology protection conjecture and argued that it is enforced by vacuum fluctuations destroying any time machine at the moment it is created (Chapter 14); made bets with Kip Thorne over whether Cygnus X-1 is a black hole (Chapter 8) and whether naked singularities can form in our Universe (Chapter 13).

Israel, Werner (b. 1931). South African born, Canadian theoretical physicist; proved that every nonspinning black hole must be spherical, and gave evidence that a black hole loses its "hair" by radiating it away (Chapter 7); discovered that the surface areas of black holes can only increase, but did not realize the significance of this conclusion (Chapter 12); with Poisson and Ori, showed that the tidal forces that surround a black hole's singularity become weaker as the hole ages (Chapter 13); developed insights into the early history of black-hole research (Chapter 3).

Kerr, Roy P. (b. 1934). New Zealander mathematician; discovered the solution to Einstein's field equation, which describes a spinning black hole: the "Kerr solution" (Chapter 7).

Landau, Lev Davidovich (1908–1968). Soviet theoretical physicist; Nobel laureate; transfused theoretical physics from Western Europe into the U.S.S.R. in the 1930s (Chapters 5 and 13); tried to explain stellar heat as produced by stellar material being captured onto a neutron core at the star's center, and thereby triggered Oppenheimer's research on neutron stars and black holes (Chapter 5); was imprisoned in Stalin's Great Terror and then released so he could develop the theory of superfluidity (Chapter 5); contributed to Soviet nuclear weapons research (Chapter 6).

Laplace, Pierre Simon (1749–1827). French natural philosopher; developed and popularized the concept of a dark star (black hole) as governed by Newton's laws of physics (Chapters 3 and 6).

Lorentz, Hendrik Antoon (1853–1928). Dutch theoretical physicist; Nobel laureate; developed key foundations for the laws of special relativity, the most important being the Lorentz–Fitzgerald contraction and time dilation (Chapter 1); friend and associate of Einstein when Einstein was developing his general relativistic laws of physics (Chapter 2).

Maxwell, James Clerk (1831–1879). British theoretical physicist; developed the laws of electricity and magnetism (Chapter 1).

Michell, John (1724–1793). British natural philosopher; developed and popularized the concept of a dark star (black hole) as governed by Newton's laws of physics (Chapters 3 and 6).